Solax Power Energy Storage Inverter
MODBUS RTU Communication protocols

Table of contents

1 Information on this Document 3
1.1 Introduction 3
1.2 Validity 3
1.3 Abbreviation 3
2 Protocols Overview 4
2.1 Protocol Type 4
2.2 Reading and Writing of Data 5
2.2.1 MODBUS Command Type 5
2.2.2 Register Address, Width and Data Block 5
2.2.3 Data Transmission 5
2.2.4 Time Request 5
2.2.5 Data Format Description 6
2.3 MODBUS Exception Responses 11
3 Use the Inverter via Commands 12
3.1 Inverter MODBUS Address Overview 12
3.2 Get Firmware Info (Function Code: 0x03) 13
3.2.1 Get Inverter Firmware Info 13
3.2.2 Get Battery Firmware Info 14
3.3 Get Inverter Real-Time Info (Function Code: 0x04) 15
3.3.1 Get Inverter Status 15
3.3.2 Get Inverter-PV Port Info 16
3.3.3 Get Inverter-Grid Port Info 17
3.3.4 Get Inverter-Battery Port Info 18
3.4 Get Inverter History Data (Function Code: 0x04) 19
3.5 Read or Set Inverter Operating Parameters 20
3.5.1 Read or Set the Inverter Work Mode Parameters 21
3.5.2 Read or Set the Inverter Charge and Discharge Period 24
3.5.3 Read or Set Inverter-Battery System Parameters 25
3.5.4 Read or Set Meter/CT Parameters 26
3.5.5 Read or Set Energy storage integrated system communication parameters 27
3.5.6 Read or Set Generator operating parameters 28
3.5.7 Read or Set the Inverter Parameters to Use the VPP function 30
3.5.8 Read or Set the Inverter Parallel System 38
3.5.9 Upgrade the Inverter via MODBUS 42
3.5.10 Reset the Inverter (Function Code: 0x06) 45
4 Inverter Run Mode Description 45

1 Information on this Document

1.1 Introduction

The MODBUS protocol is a common standard device-level communication protocol. This document describes and illustrates the MODBUS protocol for the Solax Power Energy Storage Inverter System, which is used to standardize and constrain subsequent third-party integration development. The Solax Power Energy Storage Inverter system communicates over MODBUS based serial links, which are based on RTU links and follow the MODBUS RTU format convention. This document focuses on information specific to the Solax Power Energy Storage Inverter System; for additional information, please refer to the description in the MODBUS standard specification document.

1.2 Validity

This document is valid for:

- X1-Hybrid-G4
- X3-Hybrid-G4

1.3 Abbreviation

Abbr.	Explanation
RO	Read Only
RW	Read and Write
uint8	A byte (8-bit)
uint16	A word (16-bit)
int16	A signed word (16-bit)
uint32	A double word (32-bit)
int32	A signed double word (32-bit)
MSB	Most Significant Bit
LSB	Least Significant Bit
X1	X1-Hybrid-G4
X3	X3-Hybrid-G4
ADU	Application Data Unit
PDU	Protocol Data Unit

2 Protocols Overview

2.1 Protocol Type

MODBUS RTU (**For RS485**)

Address: 1 (default)

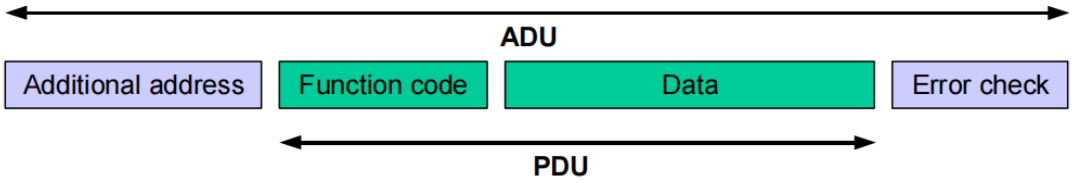
Braud Rate: 19200 (default)

Data bits: 8

Stop Bit: 1

Parity: None

Frame format:



The MODBUS protocol defines a simple protocol data unit (PDU) independent of the underlying communication layers. The mapping of MODBUS protocol on specific buses or network can introduce some additional fields on the application data unit (ADU).

Notice:
The slave address of the MODBUS RTU in the Solax Power energy storage system represents the address of the inverter. The address range is assigned as follows:

Broadcast Address	Slave Address	Reserved
0	1–247	248–255

The size of the MODBUS PDU is limited by the size constraint inherited from the first MODBUS implementation on Serial Line network (max. RS485 ADU = 256 bytes).
Therefore:
MODBUS PDU for serial line communication = 256 - Server address (1 byte) - CRC (2 bytes) = 253 bytes.
Consequently:
RS485 ADU = 253 bytes + Server address (1 byte) + CRC (2 bytes) = 256 bytes.

2.2 Reading and Writing of Data

2.2.1 MODBUS Command Type

Type	Hexadecimal Value	Data volume(number of registers)*
Read Holding Registers	0x03	1 to 125
Read Input Registers	0x04	1 to 125
Write Single Registers	0x06	1
Write Multiple Registers	0x10	1 to 123

*Number of MODBUS registers transferable as a data block per command(16 bit)

2.2.2 Register Address, Width and Data Block

A MODBUS register is 16 bits wide. For wider data items, connected MODBUS registers are used and considered as data blocks. The address of the first MODBUS register in a data block is the start address of the data block.

2.2.3 Data Transmission

With data storage in the Motorola format "Big Endian", data transmission begins with the high byte and then the low byte of the MODBUS register.

2.2.4 Time Request

Timing parameter	Value
The least interval time between two instructions	1 Sec
Character-gap time out(silent time between 2 package)	>100ms
Response timeout	1 Sec

Notice: When use "**Write Single Registers**" and "**Write Multiple Registers**" function, some registers will be write in EEprom if they are changed(these parameters can be saved after power failure). But the EEprom has the write times limit. Too frequent operation will lead to irreversible hardware damage. Related registers are marked with □. If there is any doubt about the use, please contact the technical personnel in time.

2.2.5 Data Format Description

Data format (Read Holding Register)

Master request format			
	Bytes number	Content format	Example
Slave ID	1 byte	0x00~0xFF (Inverter default 0x01)	01
Function code	1 byte	**0x03**	03
Start register address	2 byte Address MSB Address LSB	0x0000-0xFFFF	03 24
Register number	2byte Data MSB Data LSB	N	00 01
CRC	2byte CRC MSB CRC LSB		C4 45
Slave normal response			
Slave ID	1 byte	0x00~0xFF (Inverter default 0x01)	01
Function code	1 byte	**0x03**	03
Byte number	1 byte Data	2*N	02
Register data	N*2byte Data MSB Data LSB		00 00
CRC	2byte CRC MSB CRC LSB		B8 44
Slave fault response			
Slave ID	1byte	0x00~0xFF (Inverter default 0x01)	01
Fault code	1byte	**0x83**	83
Abnormal code	1byte	0x01 or 0x02 or 0x03 or 0x04	02
CRC	2byte CRC MSB CRC LSB		C5 3B

Data format (Read Input Register)

Master request format			
	Bytes number	Content format	Example
Slave ID	1 byte	0x00~0xFF (Inverter default 0x01)	01
Function code	1 byte	**0x04**	04
Start register address	2 byte Address MSB Address LSB	0x0000-0xFFFF	04 15
Register number	2byte Data MSB Data LSB	N	00 01
CRC	2byte CRC MSB CRC LSB		21 3E
Slave normal response			
Slave ID	1 byte	0x00~0xFF (Inverter default 0x01)	01
Function code	1 byte	**0x04**	04
Byte number	1 byte Data	2*N	02
Register date	N*2byte Data MSB Data LSB		00 00
CRC	2byte CRC MSB CRC LSB		B9 30
Slave fault response			
Slave ID	1byte	0x00~0xFF (Inverter default 0x01)	01
Fault code	1byte	**0x84**	84
Abnormal code	1byte	0x01 or 0x02 or 0x03 or 0x04	02
CRC	2byte CRC MSB CRC LSB		C2

Data format (Write Single Register)

Master request format			
	Bytes number	Content format	Example
Slave ID	1 byte	0x00~0xFF (Inverter default 0x01)	01
Function code	1 byte	**0x06**	06
Register address	2 byte Address MSB Address LSB	0x0000-0xFFFF	06 16
Value	2byte Data MSB Data LSB	0x0000-0xFFFF	00 01
CRC	2byte CRC MSB CRC LSB		A9 46
Slave normal response			
Slave ID	1 byte	0x00~0xFF (Inverter default 0x01)	01
Function code	1 byte	**0x06**	06
Register address	2 byte Address MSB Address LSB	0x0000-0xFFFF	06 16
Value	2byte Data MSB Data LSB	0x0000-0xFFFF	00 01
CRC	2byte CRC MSB CRC LSB		A9 46
Slave fault response			
Slave ID	1byte	0x00~0xFF (Inverter default 0x01)	01
Fault code	1byte	**0x86**	86
Abnormal code	1byte	0x01 or 0x02 or 0x03 or 0x04	
CRC	2byte CRC MSB CRC LSB		C3 A1

Data format (Write Multiple Register)

Master request format			
	Bytes number	Content format	Example
Slave ID	1 byte	0x00~0xFF (Inverter default 0x01)	01
Function code	1 byte	**0x10**	10
Register address	2 byte Address MSB Address LSB	0x0000-0xFFFF	10 00
Register number	2byte Number MSB Number LSB	0x0001-0x007B	00 07
Byte number	1Byte	2*N	0E
Value	2*N byte Data MSB Data LSB	0x0000-0xFFFF	58 42 34 30 34 30 30 30 30 30 30 30 30 30
CRC	2byte CRC MSB CRC LSB		57 EA

Slave normal response			
Slave ID	1 byte	0x00~0xFF (Inverter default 0x01)	01
Function code	1 byte	**0x10**	10
Register address	2 byte Address MSB Address LSB	0x0000-0xFFFF	10 00
Register number	2byte Number MSB Number LSB	0x0001-0x007B	00 07
CRC	2byte CRC MSB CRC LSB		85 0B
Slave fault response			
Slave ID	1byte	0x00~0xFF (Inverter default 0x01)	01
Fault code	1byte	**0x90**	90
Abnormal code	1byte	0x01 or 0x02 or 0x03 or 0x04	02
CRC	2byte CRC MSB CRC LSB		CD C1

2.3 MODBUS Exception Responses

Only some of the MODBUS exception responses are listed here, for more details please visit the MODBUS website: http://www.modbus.org.

MODBUS Exception Codes		
Code	**Name**	**Meaning**
01	IILEGAL FUNCTION	The function code received in the query is not an allowable action for the server. This may be because the function code is only applicable to newer devices, and was not implemented in the unit selected. It could also indicate that the server is in the wrong state to process a request of this type, for example because it is unconfigured and is being asked to return register values.
02	ILLEGAL DATA ADDRESS	The data address received in the query is not an allowable address for the server. More specifically, the combination of reference number and transfer length is invalid.
03	ILLEGAL DATA VALUE	A value contained in the query data field is not an allowable value for server. This indicates a fault in the structure of the remainder of a complex request, such as that the implied length is incorrect.
04	SERVER DEVICE FAILURE	Specialized use in conjunction with programming commands. The server has accepted the request and is processing it, but a long duration of time will be required to do so. This response is returned to prevent a timeout error from occurring in the client.

3 Use the Inverter via Commands

3.1 Inverter MODBUS Address Overview

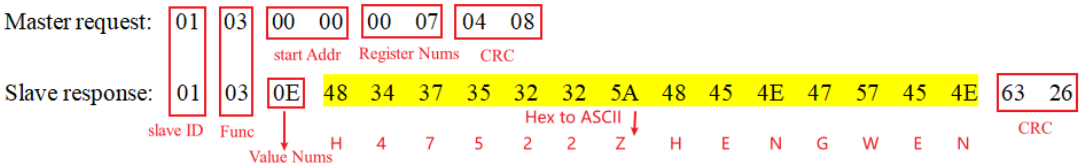
Function Code	Address	Application
0x03	0x0000~0x01FF	Inverter related configuration data or model feature information
	0x0200~0x0300	Battery characteristic information
	0x3098~0x30A9	modbus upgrade result feedback
0x04	0x0000~0x017F	Inverter real-time data
	0x0180~0x0x01DA	Italian safety self-test status and result data
	0x01DD~0x02ED	Summary of basic operating information of the inverter parallel system and slave units
0x06	All range except 0xA000-0xA1FF	Inverter configuration parameters
0x10	0x0000~0x007B	Generally write multiple parameters
	0x007C~0x008A	Vpp power control
	0x3000~0x0x30C2	modbus upgrade operation range

3.2 Get Firmware Info (Function Code: 0x03)

3.2.1 Get Inverter Firmware Info

Register	Variable	Description	Unit	Data format	Length
0x0000 ~0x0006	Inverter SN	Inverter Serial Number	-	uint16	7
0x007D	FirmwareVersion_DSP_Minor	DSP_Minor_Software_Ver	1	uint16	1
0x007E	HardwareVersion_DSP	DSP_Hardware_Version	1	uint16	1
0x007F	FirmwareVersion_DSP_Major	DSP_Major_Software_Ver	1	uint16	1
0x0080	FirmwareVersion_ARM_Major	ARM_Major_Software_Ver	1	uint16	1
0x0082	FirmwareVersion_ModbusRTU	Modbus_RTU_Software_Ver	1	uint16	1
0x0083	FirmwareVersion_ARM_Minor	ARM_Minor_Software_Ver	1	uint16	1
0x0084	FirmwareVersion_ARM_Bootloader	ARM_Bootloader_Software_Ver	1	uint16	1
0x00AA ~0x00AE	Register SN （external WiFi module）	WiFi Module SN	-	uint16	1
0x00BA	Inverter Power Type	X1: 3000/3680/5000 /6000/7500 X3: 15K/12K/10k/8K /6K/5K	1w	uint16	1
0x0105	Machine Type	1:X1 3:X3	-	uint16	1
0x0107	Machine Style	0:X-Hybrid 1:X-FIT	1	uint16	1

Example:



So the Inverter Serial Number is: H47522ZHENGWEN

3.2.2 Get Battery Firmware Info

Register	Variable	description	Unit	Data format	Length
0x0200	Subsystem_Num	Subsystem_Battery Num	1	Uint16	1
0x0201	BMS_Master Version	Version type describe x.y x = Uint8(Hi) y = Uint8(Low)	1	Uint16	1
0x0202	BMS_Slave1 Version		1	Uint16	1
0x0203	BMS_Slave2 Version		1	Uint16	1
0x0204	BMS_Slave3Version		1	Uint16	1
0x0205	BMS_Slave4Version		1	Uint16	1
0x0206	BMS_Slave5Version		1	Uint16	1
0x0207	BMS_Slave6Version		1	Uint16	1
0x0208	BMS_Slave7Version		1	Uint16	1
0x0209	BMS_Slave8Version		1	Uint16	1
0x020A~ 0x0210	Master SN	Master SN	1	14char	7
0x0211~ 0x0217	slave1_2SN	slave1_2SN	1	14char	7
0x0218~ 0x021E	slave3_4SN	slave3_4SN	1	14char	7
0x021F~ 0x0225	slave5_6SN	slave5_6SN	1	14char	7
0x0226~ 0x022C	Slave7_8SN	Slave7_8SN	1	14char	7

3.3 Get Inverter Real-Time Info (Function Code: 0x04)

3.3.1 Get Inverter Status

Register	Variable	Description	Unit	Data format	Length
0x0009	Inverter Run Mode	Inverter Run Mode Description	-	uint16	1
0x0054	Inverter Lock State	0:locked 1:unlocked	-	uint16	1
0x0040	InvFaultMessage.LSB	Please refer to the Error code in the detailed documentation.	-	uint16	1
0x0041	InvFaultMessage.MSB		-	uint16	1
0x0043	Manger Fault Message		-	uint16	1
0x0044	Bat_BMS_FaultMessage.LSB		-	uint16	1
0x0045	Bat_BMS_FaultMessage.MSB		-	uint16	1
0x0085	RTC-Seconds	RTC-Seconds	—	uint16	1
0x0086	RTC-Minutes	RTC-Minutes	—	uint16	1
0x0087	RTC-Hours	RTC-Hours	—	uint16	1
0x0088	RTC-Days	RTC-Days	—	uint16	1
0x0089	RTC-Months	RTC-Months	—	uint16	1
0x008A	RTC-Years	RTC-Years	—	uint16	1

3.3.2 Get Inverter-PV Port Info

Register	Variable	Description	Unit	Data format	Length
0x0003	PvVoltage1	PV1 Voltage	0.1V	uint16	1
0x0004	PvVoltage2	PV2 Voltage	0.1V	uint16	1
0x0005	PvCurrent1	PV1 Current	0.1A	uint16	1
0x0006	PvCurrent2	PV2 Current	0.1A	uint16	1
0x000A	Powerdc1	PV1 Power	1W	uint16	1
0x000B	Powerdc2	PV2 Power	1W	uint16	1
0x001B	MPPTCount	MPPT Count	1	uint16	1

3.3.3 Get Inverter-Grid Port Info

Register	Variable	Description	Unit	Data format	Length
0x0000	GridVoltage**(X1)**	On-Grid Voltage	0.1V	uint16	1
0x0001	GridCurrent**(X1)**	On-Grid Current	0.1A	int16	1
0x0002	GridPower**(X1)**	On-Grid Power	1W	int16	1
0x0007	GridFrequency**(X1)**	On-Grid Frequency	0.01Hz	uint16	1
0x001A	Grid Status	0: On Grid 1: Off Grid	-	uint16	1
0x0088 ~0x0089	On-grid Run Time	On-grid Run Time (0x88:LSB,0x89:MSB)	0.1h	int32	2
0x008A ~0x008B	Off-grid Run Time	Off-gridRunTime (0x8A:LSB,0x8B:MSB)	0.1h	int32	2
0x004C	Off-grid Voltage**(X1)**	Off-grid Voltage	0.1V	uint16	1
0x004D	Off-grid Current**(X1)**	Off-grid Current	0.1A	uint16	1
0x004E	Off-grid Power**(X1)**	Off-grid power	1VA	uint16	1
0x004F	Off-grid Frequency**(X1)**	Off-grid _Frequency	0.01Hz	uint16	1

3.3.4 Get Inverter-Battery Port Info

Register	Variable	Description	Unit	Data format	Length
0x0017	BMS_Connect_State	0:Disconnected 1:Connected	-	uint16	1
0x0018	Bat Temperature	BAT Temperature	1℃	int16	1
0x0014	BatVoltage_Charge1	BatVoltage_Charge1	0.1V	int16	1
0x0015	BatCurrent_Charge1	BatCurrent_Charge1	0.1A	int16	1
0x0016	Batpower_Charge1	Batpower_Charge1	1W	int16	1
0x0019	BDCStatus	0: discharge 1: charge 2: stop	-	uint16	1
0x001C	Battery Capacity	Battery capacity	1%	uint16	1
0x0024	BMS ChargeMaxCurrent	BMS ChargeMaxCurrent	0.1A	uint16	1
0x0025	BMS DischargeMaxCurrent	BMS DischargeMaxCurrent	0.1A	uint16	1
0x0026 ~0x0027	BMS_BatteryCapacity	BMS_BatteryCapacity	Wh	uint16	1
0x00BA	Battery_Tem_High	Battery_Tem_High	0.1℃	int16	1
0x00BB	Battery_Tem_Low	Battery_Tem_Low	0.1℃	int16	1
0x00BC	Cell_Voltage_High	Cell_Voltage_High	0.001V	Uint16	1
0x00BD	Cell_Voltage_Low	Cell_Voltage_Low	0.001V	Uint16	1
0x00BE	BMS_UserSOC	BMS_UserSOC	1%	Uint16	1
0x00BF	BMS_UserSOH	BMS_UserSOH	1%	Uint16	1

3.4 Get Inverter History Data (Function Code: 0x04)

	Register	Variable	Description	Unit	Data format	Length
On-Grid	0x0050	Etoday_togrid	Today Energy (Inverter AC Port)	0.1kWh	uint16	1
	0x0052 ~0x0053	Etotal_togrid	Total Energy (Inverter AC Port) (0x52:LSB,0x53:MSB)	0.1kWh	uint32	2
	0x0091	EchargeToday	EchargeToday (Inverter AC Port)	0.1kWh	uint16	1
	0x0092 ~0x0093	EchargeTotal	EchargeTotal (Inverter AC Port) (0x92:LSB,0x93:MSB)	0.1kWh	uint32	2
EPS (Off-Grid)	0x008E ~0x008F	Off-gridYieldTotal	Off-grid Yield Total (0x8E:LSB,0x8F:MSB)	0.1kWh	uint32	2
	0x0090	Off-gridYieldToday	Off-grid Yield Today	0.1kWh	uint16	1
Meter /CT_1	0x0048 ~0x0049	feedin_energy_total(meter)	energy to the grid (0x48:LSB,0x49:MSB)	0.01kWh	uint32	2
	0x004A ~0x004B	consum_energy_total(meter)	energy form the grid (0x4A:LSB,0x4B:MSB)	0.01kWh	uint32	2
	0x0098 ~0x0099	feedin_energy_today	energy to the grid (meter) (0x98:LSB,0x99:MSB)	0.01kWh	uint32	2
	0x009A ~0x009B	consum_energy_today	energy form the grid (meter) (0x9A:LSB,0x9B:MSB)	0.01kWh	uint16	1
Meter /CT_2	0x00AC ~0x00AD	consum_energy_total_Meter2	energy form the grid (0xAC:LSB,0xAD:MSB)	0.01kWh	uint32	2
	0x00B0 ~0x00B1	consum_energy_today_Meter 2	energy form the grid (0xB0:LSB,0xB1:MSB)	0.01kWh	uint16	1

3.5 Read or Set Inverter Operating Parameters

Write Register (0x06)	Read Register (0x03)	Variable	Description	Unit	Data format	Length
0x0000	**-**	Unlock Password	Unlock Password	1	uint16	1
0x001C	**0x00DA**	Inverter OutPut_Switch	1=ON; 0=Off	0/1	uint16	1
0x0039	**0x00E0**	User Password	User Password 0000~9999	1	uint16	1
0x003A	**0x00E1**	Advance Password	Advance Password 0000~9999	1	uint16	1
0x0047	**0x00BB**	Language	0:English 1:German 2: French 3: Polish 4:Spanish 5:Portuguese 6:Italian 7:Chinese(Mute) 8: Ukrainian 9: Brazil	0~9	uint16	1
0x0000	**0x001D**	Safety Code	Please refer to the Table of Safety code in the detailed documentation.	1	uint16	1

3.5.1 Read or Set the Inverter Work Mode Parameters

Work Mode	Write Register (0x06)	Read Register (0x03)	Variable	Description	Unit	Data format	Length
-	0x001F	0x008B	Inverter Work Mode	0:Self use mode 1: Feedin Priority 2:Back up mode 3:Manual mode 4: Peak Shaving 5：TOU Mode 6: Smart Schedule	1	uint16	1
Self Use	0x0061	0x0093 (Hi)	SelfUse_Discharge_MinSoC	10%~100%	1%	Uint8(Hi)	1
	0x0062	0x0093(Lo)	SelfUse_NightCharge_Enable	Whether to allow electricity from the grid 0:Disable 1:Enable	1	Uint8(Lo)	
	0x0063	0x0094	SelfUse_NightCharge_UpperSoC	This value will be enabled if SelfUse_NightCharge_Enable is 1. 10%~100%	1%	Uint16	1
Feedin Priority	0x0065	0x0095(Hi)	Feedin_NightCharge_UpperSoC	10%~100%	1%	uint8(Hi)	1
	0x0064	0x0095(Lo)	Feedin_Discharge_MinSoC	10%~100%	1%	uint8(Lo)	
Backup Mode	0x0067	0x0096(Lo)	BackUp_Discharge_MinSoC	30%~100%	1%	uint8(Lo)	1
	0x0066	0x0096(Hi)	BackUp_NightCharge_UpperSoC	30%~100%	1%	uint8(Hi)	
Manual	0x0020	0x008C	Manual mode	0:Stop charge&discharge 1:Force charge 2:Force discharge	1	uint16	1
Peak Shaving Mode	0x00EA(Hi)	0x014F(Hi)	PeakShaving DischarPeriod. bP1_StartMinute	0-59	1min	uint8(Hi)	1
	0x00EA(Lo)	0x014F(Lo)	PeakShaving DischarPeriod.	0-23	1hour	uint8(Lo)	

			bP1_StartHour				
	0x00EB(Hi)	**0x0150(Hi)**	PeakShaving DischarPeriod. bP1_StopMinute	0-59	1min	uint8(Hi)	1
	0x00EB(Lo)	**0x0150(Lo)**	PeakShaving DischarPeriod. bP1_StopHour	0-23	1hour	uint8(Lo)	
	0x00EE	**0x0153**	PeakShaving. PeriodB PeakLimits1	Peak Shaving Mode Discharge Period 1 Power Limit	1W	Uint16	1
	0x00EC(Hi)	**0x0151(Hi)**	PeakShavingDischarPeriod.bP2_StartMinute	0-59	1min	uint8(Hi)	1
	0x00EC(Lo)	**0x0151(Lo)**	PeakShavingDischarPeriod.bP2_StartHour	0-23	1hour	uint8(Lo)	
	0x00ED(Hi)	**0x0152(Hi)**	PeakShavingDischarPeriod.bP2_StopMinute	0-59	1min	uint8(Hi)	1
	0x00ED(Lo)	**0x0152(Lo)**	PeakShavingDischarPeriod.bP2_StopHour	0-23	1hour	uint8(Lo)	
	0x00EF	**0x0154**	PeakShaving. PeriodDPeakLimits2	Peak Shaving Mode Discharge Period 2 Power Limit	1w	Uint16	1
	0x00F0	**0x0155**	PeakShaving. PeriodAChargeFromGridEn	From Grid charging switch	1	Uint16	1
	0x00F1	**0x0156**	PeakShaving .Period AChargePowerLimits	Charging power value from grid	1W	Uint16	1
	0x00F2	**0x0157**	PeakShaving .Period AMax_SOC	Maximum SOC charged from grid	1%	Uint16	1
	0x00F3	**0x0158**	PeakShaving .PeriodC Reserved_SOC	Peak shaving mode reserved SOC	1%	uint16	1
TOU Mode	-	**0x016E**	TOUMode_TotalMinSoc	10~100	1%	uint8(Lo)	1
			TOUMode_WorkMode	0xA0：SelfUse 0xA1：AllowCharging	1	uint8(Hi)	

				0xA2：ForceDischarging 0xA3：BatteryOff 0xA4：PeakShaving			
		0x016F	TOUMode_SelfuseMinSOC	10~100	1%	uint16	1
		0x0170	TOUMode_ChargeFromGridEn	0xA0:Disable 0xA1:Enable	1	uint8(Lo)	1
			TOUMode_ChargeStopSOC	10~100	1%	uint8(Hi)	
		0x0171	TOUMode_DischgPowerLimitRate	0~100	1%	uint8(Lo)	1
			TOUMode_DischargeMinSOC	10~100	1%	uint8(Hi)	
		0x0172 ~0x0173	TOUMode_PeakShavingLimit	-	1w	Uint32	2
Smart Schedule	**0x00FF**	**0x0182**	Smart Schedule Work Mode	0: Self Use 1: Feedin Priority 2: BAT not Discharge	1	uint16	1

3.5.2 Read or Set the Inverter Charge and Discharge Period

Write Register (0x06)	Read Register (0x03)	Variable	Description	Unit	Data format	Length
0x0068	0x0097	ChargePeriod1_StartMinute	0-59	1M	uint8(Hi)	1
		ChargePeriod1_StartHour	0-23	1H	uint8(Lo)	
0x0069	0x0098	ChargePeriod1_EndMinute	0-59	1M	uint8(Hi)	1
		ChargePeriod1_EndHour	0-23	1H	uint8(Lo)	
0x006A	0x0099	DischargePeriod1_StartMinute	0-59	1M	uint8(Hi)	1
		DischargePeriod1_StartHour	0-23	1H	uint8(Lo)	
0x006B	0x009A	DischargePeriod1_EndMinute	0-59	1M	uint8(Hi)	1
		DischargePeriod1_EndHour	0-23	1H	uint8(Lo)	
0x006C	0x009B	Set_Chrg&DischrgPeriod2_Enable	Whether to use period 2. 0:Disable 1:Enable	1	uint16	1
0x006D	0x009C	ChargePeriod2_StartMinute	0-59	1M	uint8(Hi)	1
		ChargePeriod2_StartHour	0-23	1H	uint8(Lo)	
0x006E	0x009D	ChargePeriod2_EndMinute	0-59	1M	uint8(Hi)	1
		ChargePeriod2_EndHour	0-23	1H	uint8(Lo)	
0x006F	0x009E	DischargePeriod2_StartMinute	0-59	1M	uint8(Hi)	1
		DischargePeriod2_StartHour	0-23	1H	uint8(Lo)	
0x0070	0x009F	DischargePeriod2_EndMinute	0-59	1M	uint8(Hi)	1
		Discharge Period2_EndHour	0-23	1H	uint8(Lo)	

3.5.3 Read or Set Inverter-Battery System Parameters

Write Register (0x06)	Read Register (0x03)	Variable	Description	Unit	Data format	Length
0x0021	0x008D	Battery Type	0：Lead Acid 1：Lithium	1	uint16	1
0x0024	0x0090	Battery_ChargeMaxCurrent	Lithium battery charge maximum current	0.1A	uint16	1
0x00E0	0x010E	BatteryChargeMaxSoc	Lithium battery Charger upper limit	1%	uint16	1
0x00CF	0x00A3	BatteryHeatingEn	0:disable 1:enable	-	uint16	1
0x0D0	0x00A4	HeatingPeriod1_StartMinute	0-59	1M	uint8(Hi)	1
		HeatingPeriod1_StartHour	0-23	1H	uint8(Lo)	
0x00D1	0x00A5	HeatingPeriod1_EndMinute	0-59	1M	uint8(Hi)	1
		HeatingPeriod1_EndHour	0-23	1H	uint8(Lo)	
0x00D2	0x00A6	HeatingPeriod2_StartMinute	0-59	1M	uint8(Hi)	1
		HeatingPeriod2_StartHour	0-23	1H	uint8(Lo)	
0x00D3	0x00A7	HeatingPeriod2_EndMinute	0-59	1M	uint8(Hi)	1
		HeatingPeriod2_EndHour	0-23	1H	uint8(Lo)	
0x00E1	0x010F	bBatteryChargeToEVC	0:Enable1:Disable	1	uint16	1
0x009A	0x00A2	Extend Bms Setting	0:disable 1:enable	1	uint16	1

3.5.4 Read or Set Meter/CT Parameters

Write Register (0x06)	Read Register (0x03)	Variable	Description	Unit	Data format	Length
0x00AA	**0x0115**	Meter/CT_Select	0:Meter 1:CT	1	uint16	1
0x00A1	**0x0109**	Meter1_ID	Meter1 ID 1~200	1	uint16	1
0x00A2	**0x010A**	Meter2_ID	Meter2 ID 1~200	1	uint16	1
0x00A4	**0x010B**	DirectionMeterCT1	0:Positive 1:Negative	1	uint16	1
0x00A5	**0x010C**	DirectionMeterCT2	0:Positive 1:Negative	1	uint16	1
0x00F5	**0x00B3**	FastCtCheckEn	0:disable 1:enable	1	uint16	1

3.5.5 Read or Set Energy storage integrated system communication parameters

Write Register (0x06)	Read Register (0x03)	Variable	Description	Unit	Data format	Length
0x00C9	**0x00AF**	ModBusRTU_Address	ModBusRTU_Address	1	uint16	1
0x00CA	**0x00B0**	ModBusRTU_BraudRate	0:115200 1:57600 2:56000 3:38400 4:19200 5:14400 6:9600	bit/s	uint16	1
0x008F	**0x013E**	485CommFunSelect	0:modbus 485 1:EV Charger 2: DadaHub 3:AdatptBoxG2 4: EVC& AdaptBoxG2 5: AdaptBoxG2 & Meter	1	uint16	1
0x00F9	**0x015C**	EVChargerAddr	0~255	1	uint16	1
0x00FB	**0x015E**	AdaptBoxG2Addr	0~255	1	uint16	1

3.5.6 Read or Set Generator operating parameters

	Write Register (0x06)	Read Register (0x03)	Variable	Description	Unit	Data format	Length
-	0x00C7	0x0131	External GeneratorEn	0:Disable 1:ATS Control 2:Dry Contact	1	uint16	1
ATS &Dry Contact	0x00C8	0x0132	ExternalGen MaxCharge	ExternalGenMaxCharge	X1: 1W X3: 10W	uint16	1
	0x0109	0x016B	ChargeFrom GenEnable	0:Disable 1:Enable	1	uint16	1
	0x010A	0x016C	ChargeFromGen_ChargeSoC	10~100	1%	uint16	1
	0x010B	0x0148	GenMinPower	0~60000	1w	uint16	1
Only Dry Contact	0x00E3	0x0140	Start Gen Method	0:reference soc 1:immediately	1	uint16	1
	0x00E4	0x0141	Switch on SoC	Switch on SoC(reference soc)	1%	uint16	1
	0x00E5	0x0142	Switch off SoC	Switch off SoC(reference soc)	1%	uint16	1
	0x00E6	0x0143	MaxRunTime	MaxRunTime (1~60000)	1Min	uint16	1
	0x00E7	0x0145	MinRestTime	MinRestTime (1~60000)	1Min	uint16	1
	0x00E8	0x0146	Allow Work start time Minute	Allow Work start time Minute	1M	uint8(Hi)	1
			Allow Work start time Hour	Allow Work start time Hour	1H	uint8(Lo)	1
	0x00E9	0x0147	Allow Work stop time Minute	Allow Work start time Minute	1M	uint8(Hi)	1
			Allow Work stop time Hour	Allow Work start time Hour	1H	uint8(Lo)	1
Work Period	0x0100	0x0162	GenCharge_StartHour	0-23	1	uint8(Hi)	1
			GenCharge_StartMinute	0-59	1	uint8(Lo)	1
	0x0101	0x0163	GenCharge_EndHour	0-23	1	uint8(Hi)	1
			GenCharge_EndMinute	0-59	1	uint8(Lo)	1

	0x0102	**0x0164**	GenDischarge_StartHour	0-23	1	uint8(Hi)	1
			GenDischarge_StartMinute	0-59	1	uint8(Lo)	1
	0x0103	**0x0165**	GenDischarge_EndHour	0-23	1	uint8(Hi)	1
			GenDischarge_EndMinute	0-59	1	uint8(Lo)	1
	0x0104	**0x0166**	GenP2_SetEnable	0:Disable 1:Enable	1	uint16	1
	0x0105	**0x0167**	GenP2Charge_StartHour	0-23	1	uint8(Hi)	1
			GenP2Charge_StartMinute	0-59	1	uint8(Lo)	1
	0x0106	**0x0168**	GenP2Charge_EndHour	0-23	1	uint8(Hi)	1
			GenP2Charge_EndMinute	0-59	1	uint8(Lo)	1
	0x0107	**0x0169**	GenP2Discharge_StartHour	0-23	1	uint8(Hi)	1
			GenP2Discharge_StartMinute	0-59	1	uint8(Lo)	1
	0x0108	**0x016A**	GenP2Discharge_EndHour	0-23	1	uint8(Hi)	1
			GenP2Discharge_EndMinute	0-59	1	uint8(Lo)	1

3.5.7 Read or Set the Inverter Parameters to Use the VPP function

(Function Code: 0x10 (old))

Register	Variable	Description	Unit	Data format	Length
0x007C	Modbus Power Control	0:disable remote control 1:enable power control 2:enable electric quantity control 3:enable SOC target control 4:enable push power - positive/negative mode 5:enable push power -zero mode 6:enable self consume Charge-Discharge Mode 7:enable self consume Charge Only Mode	1	uint16	1
0x007D	Target Set Type	1：set 2：update	1	uint16	1
0x007E ~0x007F	RemoteControl ActivePower	0x007E(LSB) 0x007F(MSB) （Postive mean charge; Negative mean discharge)	1W	int32	2
0x0080 ~0x0081	RemoteControl ReactivePower	0x0080(LSB) 0x0081(MSB) (Positive mean Inductive reactive power;Negative mean Capacitive reactive power)	1Var	int32	2
0x0082	Time_of_Duration	power control mode Time of Duration	1s	Uint16	1
0x0083	TargetSoc	Target soc	1%	Uint16	1
0x0084 ~0x0085	TargetEnergy	0x0084(LSB) 0x0085(MSB)	1Wh	Uint32	2

0x0086 ~0x0087	Charge_Discharg_Power	0x0086(LSB) 0x0087(MSB) The power of charging or discharging （Postive mean charge; Negative mean discharge)	1W	Int32	2
0x0088	RemoteCtrlTimeOut	Timeout counter	1s	Uint16	1
0x0089 ~0x008A	PushModePower	0x0089(LSB) 0x008A(MSB) The power of charging or discharging （Postive mean discharge Negative mean charge)	1W	int32	2

(Function Code: 0x10 (new))

Register	Variable	Description	Unit	Data format	Length
0x00A0	Power Control Mode	0:disable remote control 1:enable power control 2:enable electric quantity control 3:enable SOC target control 4:enable push power - positive/negative mode 5:enable push power -zero mode 6:enable self consume Charge-Discharge Mode 7:enable self consume Charge Only Mode 8:enable electrical energy control mode 9:enable electrical energy control mode，based on target capacity 11:only for mode switching 12:only for mode switching	1	uint16	1
While PowerControlMode == 0 （VPP Mode Disable）					

While PowerControlMode == 1 （enable power control）					
0xA1	TargetSetType	1: Set Target (Switch modes immediately) 2: Update Target (Switching to a new mode after execution of the current operating mode)	1	Uint16	1
0xA2	ActivePower(LSB)	（Positive mean charge; Negative mean discharge)	1W	Uint16	1
0xA3	ActivePower(HSB)		1W	Uint16	1
0xA4	ReactivePower(LSB)	（Positive mean charge; Negative mean discharge)	1W	Uint16	1
0xA5	ReactivePower(HSB)		1W	Uint16	1
0xA6	Time of Duration	Time of Duration	1s	Uint16	1
0xA7	RemoteControlTimeOut	Remote Control Timeout	1W	Uint16	1
While PowerControlMode == 2 （electric quantity control）					
0xA1	TargetSetType	1: Set Target (Switch modes immediately) 2: Update Target (Switching to a new mode after execution of the current operating mode)	1	Uint16	1

0xA2	TargetEnergy(LSB)	Target electrical energy	1	Uint16	1
0xA3	TargetEnergy(HSB)		1	Uint16	1
0xA4	Charge_Discharg_Power(LSB)	The power of charging or discharging （Postive mean charge; Negative mean discharge)	1	Uint16	1
0xA5	Charge_Discharg_Power(HSB)		1	Uint16	1
0xA6	RemoteControlTimeOut	Remote Control Timeout	1s	Uint16	1
While PowerControlMode == 3 （SOC target control）					
0xA1	TargetSetType	1: Set Target (Switch modes immediately) 2: Update Target (Switching to a new mode after execution of the current operating mode)	1	Uint16	1
0xA2	TargetSoc	Target soc	1%	Uint16	1

0xA3	Charge_Discharg_Power(LSB)	The power of charging or discharging （Postive mean charge; Negative mean discharge)	1	Uint16	1
0xA4	Charge_Discharg_Power(HSB)	Remote Control Timeout	1	Uint16	1
0xA5	RemoteControlTimeOut	1: Set Target (Switch modes immediately) 2: Update Target (Switching to a new mode after execution of the current operating mode)	1s	Uint16	1
While PowerControlMode == 4 （Push Power -Positive/Negative Mode）					
0xA1	PushModePower(LSB)	Battery power target (positive discharge, negative charge)	1W	Uint16	1
0xA2	PushModePower(MSB)		1W	Uint16	1
0xA3	RemoteCtrlTimeOut	Remote Control Timeout	1s	Uint16	1
0xA4	NextMotion	Mode to be executed by the device after remote control timeout 0xA0: VPP OFF 0xA1:VPP Mode 6	1	Uint16	1

While PowerControlMode == 5 （Push Power -Zero Mode）					
0xA1	RemoteCtrlTimeOut	Remote Control Timeout	1s	Uint16	1
0xA2	NextMotion	Mode to be executed by the device after remote control timeout 0xA0: VPP OFF 0xA1:VPP Mode 6	1	Uint16	1
While PowerControlMode == 6 （Self Consume　Charge-Discharge Mode）					
0xA1	RemoteCtrlTimeOut	Remote Control Timeout	1s	Uint16	1
0xA2	NextMotion	Mode to be executed by the device after remote control timeout 0xA0: VPP OFF 0xA1: VPP Mode 6	1	Uint16	1
While PowerControlMode == 7 （Self Consume　Charge Only Mode）					
0xA1	RemoteCtrlTimeOut	Remote Control Timeout	1s	Uint16	1

0xA2	NextMotion	Mode to be executed by the device after remote control timeout 0xA0: VPP OFF 0xA1:VPP Mode 6	1	Uint16	1
While PowerControlMode == 8 （Electrical energy control mode）					
0xA1	TargetSetType	1: Set Target (Switch modes immediately) 2: Update Target (Switching to a new mode after execution of the current operating mode)	1	Uint16	1
0xA2	PVPowerLimit(LSB)	LSB of PV Power Limit	1W	Uint16	1
0xA3	PVPowerLimit(MSB)	MSB of PV Power Limit	1W	Uint16	1
0xA4	PushModePower(LSB)	Battery power target (positive discharge, negative charge) LSB of PushModePower	1W	Uint16	1
0xA5	PushModePower(MSB)	MSB of PushModePower	1W	Uint16	1
0xA6	RemoteControlTimeOut	Remote Control Timeout	1s	Uint16	1

0xA7	Next_Motion (When TimeOut)	远程控制超时后需设备执行的模式 0xA0: VPP OFF 0xA1:Default VPP Mode	1	Uint16	1
While PowerControlMode == 9 （Electrical energy control mode，based on target capacity）					
0xA1	TargetSetType	1: Set Target 2: Update Target	1	Uint16	1
0xA2	PVPowerLimit(LSB)	LSB of PV Power Limit	1W	Uint16	1
0xA3	PVPowerLimit(MSB)	MSB of PV Power Limit	1W	Uint16	1
0xA4	PushModePower(LSB)	Battery power target LSB of PushModePower	1W	Uint16	1
0xA5	PushModePower(MSB)	MSB of PushModePower	1W	Uint16	1
0xA6	Target SOC	Target SOC (Range 0~100%)	1%	Uint16	1

0xA7	RemoteControlTimeOut	Remote Control Timeout	1s	Uint16	1
0xA8	Next_Motion (When TimeOut)	远程控制超时后需设备执行的模式 0xA0: VPP OFF 0xA1:Default VPP Mode	1	Uint16	1
While PowerControlMode == 11 (only for mode switching)					
While PowerControlMode == 12 (only for mode switching)					

3.5.8 Read or Set the Inverter Parallel System

Set the Inverter to support Parallel System:

Write Register (0x06)	Read Register (0x03)	Variable	Description	Unit	Data format	Length
0x00C6	**0x0130**	Parallel Setting	0:Free 1: Master 2:Slave	1	uint16	1

Read the Parallel System Real Time Data(Take two inverters as an example):

Register	Variable	Description	Unit	Data format	Length
0x01DD	SystemInvNum	SystemInvNum	1	uint16	1
0x01E0 ~0x01E1	InvActivePower_R_All	InvActivePower_R_All	1W	int32	2

0x01E2 ~0x01E3	InvActivePower_S_All	InvActivePower_S_All	1W	int32	2
0x01E4 ~0x01E5	InvActivePower_T_All	InvActivePower_T_All	1W	int32	2
0x01E6 ~0x01E7	InvReactiveOrApparentPower_R_All	InvReactiveOrApparentPower_R_All	1VA	int32	2
0x01E8 ~0x01E9	InvReactiveOrApparentPower_S_All	InvReactiveOrApparentPower_S_All	1VA	int32	2
0x01EA ~0x01EB	InvReactiveOrApparentPower_T_All	InvReactiveOrApparentPower_T_All	1VA	int32	2
0x01EC ~0x01ED	InvCurrent_R_All	InvCurrent_R_All	0.1A	int32	2
0x01EE ~0x01EF	InvCurrent_S_All	InvCurrent_S_All	0.1A	int32	2
0x01F0 ~0x01F1	InvCurrent_T_All	InvCurrent_T_All	0.1A	int32	2
0x01F2 ~0x01F3	PvPower_ChannelA_All	PvPower_ChannelA_All	1W	uint32	2
0x01F4 ~0x01F5	PvPower_ChannelB_All	PvPower_ChannelB_All	1W	uint32	2
0x01F6 ~0x01F7	PvCurrent_ChannelA_All	PvCurrent_ChannelA_All	0.1A	uint32	2
0x01F8 ~0x01F9	PvCurrent_ChannelB_All	PvCurrent_ChannelB_All	0.1A	uint32	2

0x01FA ~0x01FB	BatPower_All	BatPower_All	1W	int32	2
0x01FC ~0x01FD	BatCurrent_All	BatCurrent_All	0.1A	int32	2
0x01FE ~0x01FF	ChargePowerLimit_All	ChargePowerLimit_All	1W	int32	2
0x0200 ~0x0201	DischargePowerLimit_All	DischargePowerLimit_All	1W	int32	2
0x0204	InvActivePower_R	Slaver 1 data	1W	int16	1
0x0205	InvActivePower_S		1W	int16	1
0x0206	InvActivePower_T		1W	int16	1
0x0207	InvReactiveOrApparentPower_R		1VA	int16	1
0x0208	InvReactiveOrApparentPower_S		1VA	int16	1
0x0209	InvReactiveOrApparentPower_T		1VA	int16	1
0x020A	InvCurrent_R		0.1A	int16	1
0x020B	InvCurrent_S		0.1A	int16	1
0x020C	InvCurrent_T		0.1A	int16	1
0x020D	PvPower_ChannelA		1W	uint16	1
0x020E	PvPower_ChannelB		1W	uint16	1
0x020F	PvVoltage_ChannelA		0.1V	uint16	1
0x0210	PvVoltage_ChannelB		0.1V	uint16	1
0x0211	PvCurrent_ChannelA		0.1A	uint16	1
0x0212	PvCurrent_ChannelB		0.1A	uint6	1
0x0213	BatPower		1W	uint16	1
0x0214	BatVoltage		0.1V	uint16	1
0x0215	BatCurrent		0.1A	uint16	1
0x0216	ChargePowerLimit		1W	uint16	1
0x0217	DischargePowerLimit		1W	uint16	1
0x0218	BatFaultMessage		1	uint16	1
0x0219	BatCapacity		1%	uint16	1

3.5.9 Upgrade the Inverter via MODBUS

Function Code	Update W/R register						
	Register	Variable	W/R	Description	Unit	Data Format	Length
0x03	0x3000 ~0x3001	BootloaderVersion	R	Bootloader Version	-	uint16	2
0x03/0x10	0x3002	IAP_Protocol	WR	bit0:data transfer protocol bit1:high power upgrade protocol	-	uint16	1
	0x3003	UpgradeModule	WR	0：Rev 1：ARM 2：MDSP 3：SDSP 4：ARC 5：ARM+DSP 6：BMS_M 7: BMS_S	-	uint16	1
	0x3004	UpgradeTimeOut	WR	Upgrade TimeOut	1S	uint16	1
	0x3005 ~0x3006	UpgradeKey	WR	Upgrade Key	-	uint16	2
	0x3007 ~0x3008	UpgradeSeed	R	UpgradeSeed	-	uint16	2
	0x3010	UpgradeMachineType	WR	UpgradeMachineType	-	uint16	1
	0x3011 ~0x3012	FileCheckSum	WR	FileCheckSum	-	uint16	2
	0x3013	DownLoadBlockNum	WR	data transfer mode:1 high power transfer mode:DownLoadBlockNum	-	uint16	1
	0x3014 ~0x3015	EraseStartAddr	WR	EraseStartAddr	-	uint16	2
	0x3016 ~0x3017	EraseLength	WR	EraseLength	-	uint16	2
	0x3018 ~0x3019	BlockStartAddr	WR	BlockStartAddr	-	uint16	2
	0x301A ~0x301B	BlockLength	WR	BlockLength	-	uint16	2
	0x301C	CurrentBlockNum	WR	data transfer mode:1 high power transfer mode:CurrentBlockNum	-	uint16	1

	0x301D ~0x301E	BlockCheckSum	WR	BlockCheckSum	-	uint16	2
	0x301F	UpgradeDataPackageNum	WR	UpgradeDataPackageNum	-	uint16	1
	0x3020 ~0x3097	UpgradeData	WR	UpgradeData	-	uint16	120
	0x3018 ~0x3019	BlockStartAddr	WR	BlockStartAddr	-	uint16	2
	0x301A ~0x301B	BlockLength	WR	BlockLength	-	uint16	2
	0x301C	CurrentBlockNum	WR	data transfer mode:1 high power transfer mode:CurrentBlockNum	-	uint16	1
	0x301D ~0x301E	BlockCheckSum	WR	BlockCheckSum	-	uint16	2
	0x301F	UpgradeDataPackageNum	WR	UpgradeDataPackageNum	-	uint16	1
	0x3020 ~0x3097	UpgradeData	WR	UpgradeData	-	uint16	120
	0x3098	BlockCheckResult	R	BlockCheckResult	-	uint16	1
	0x3099	McuDownLoadCheckResult	R	McuDownLoadCheckResult	-	uint16	1
	0x30A4	ToBeDownloadMcuInfor	R	ToBeDownloadMcuInfor	-	uint16	1
	0x30A5	DownloadedMcuInfor	R	DownloadedMcuInfor	-	uint16	1
	0x30A6	UpgradeMcuInfor	R	UpdateMcuInfor	-	uint16	1
	0x30A7	IapState	R	0x0000:AppCommonRunStatus 0x0001:AppResumeWaitStatus 0x0002:EraseProgramStatus 0x0003:ProgramDownloadStatus 0x0004:UpgradeSuccessStatus 0x0005:UpgradeFailStatus 0x8000:bootloaderCommonRunStatus 0x8001:BootloaderResumeWaitStatus	-	uint16	1
	0x30A8	DownloadedBlockNum	R	DownloadedBlockNum	-	uint16	1
	0x30A9	DownloadedPackageNum	R	DownloadedPackageNum	-	uint16	1

	0x30AA ~0x30C2	File_Name	W R	File_Name	-	uint16	25

3.5.10 Reset the Inverter (Function Code: 0x06)

Register	Variable	Description	Unit	Data format	length
0x001D	Factory Reset	Write 1 effect	1	unt16	1
0x005F	Reset_Mgr_EE	1:Reset normal configuration.	1	uint16	1
0x00A3	Reset Meter2 Energy	Write 1 effect	1	uint16	1
0x00DB	Reset Error Log	Write 1 effect	-	uint16	1
0x00DC	Reset INV Energy	Write 1 effect	-	uint16	1
0x00DF	Reset INV	Write 1 effect	-	uint16	1

4 Inverter Run Mode Description

Run mode	
Code	**Description**
0	Waiting
1	Checking
2	Normal
3	Fault
4	Permanent Fault
5	Update
6	Off-grid waiting
7	Off-grid
8	Self Testing
9	Idle
10	Standby
11~19	REV
20	NormalMode(R)
21	NormalMode(TOU-S)
22	NormalMode(TOU-C)
23	NormalMode(TOU-D)

24	NormalMode(TOU-B)
25	NormalMode(TOU-P)
26	NormalMode(Gen)
27	WaitChargeMode
28	NormalMode(SS)

For more details, please refer to the document
《(Solax)Hybrid X1&X3-G4 Modbus RTU Detailed.xlsx》,
《(Solax)Hybrid X1-G4 Inverter Function Overview.xmind》